



ABCfold NPF Conformation Pipeline

Six structure-prediction backends, launched together, clustered by TM-helix alignment

Etienne Reboul

Groupe Gibbérellines et adaptation à l'environnement
Institut de Biologie Moléculaire des Plantes (IBMP), Strasbourg

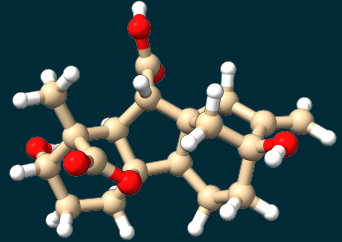
August 17, 2026

Outline

1. Background
2. Motivation
3. Pipeline overview
4. Pre-processing
5. Processing
6. Post-processing
7. Status & next steps

Gibberellins

- Key phytohormone in plants; isolated in 1931 from *Gibberella fujikuroi* (fungus), later found in plants (1950)
- Promotes growth and development:
 - Stem elongation
 - Seed germination
 - Flowering

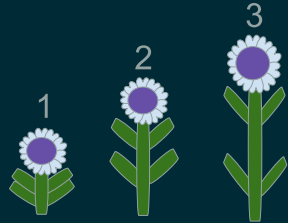


Gibberellin A1 – $C_{19}H_{24}O_6$, 348.4 g/mol
(PubChem CID 5280379)

Gibberellins – growth effect

- GA-deficiency mutants (key oxidase disrupted):
 - Smaller but more resilient
 - **Higher grain yield**

⇒ GA-deficiency-linked yield gains underpinned the **Green Revolution** (1970s)

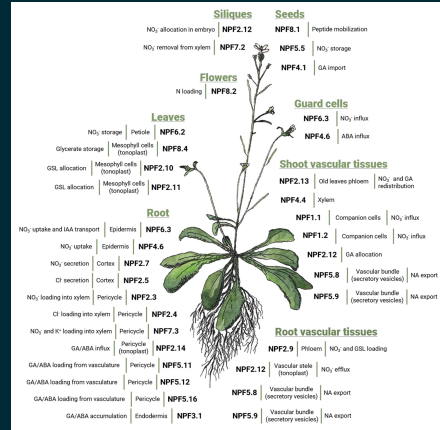


(1) No gibberellins: dwarf, no internode elongation. (2) Average gibberellins: average internode length. (3) High gibberellins: long internode length, more stem cell division.

Lee Spahr, *The effect of Gibberellins*, CC BY-SA 3.0, via Wikimedia Commons
(recolored for contrast)

Nitrate and Peptide Transporters (NPF)

- Part of the **Major Facilitator Superfamily (MFS)** of membrane transporters
- Transport:
 - Nitrate (NO_3^-)
 - Dipeptides
 - Gibberellins
- Specific to tissue and substrate



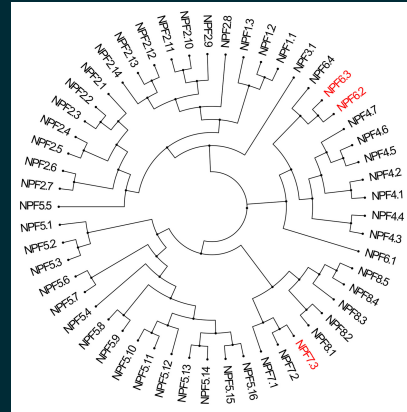
Tissue localization and function of characterized NPFs in *Arabidopsis thaliana*; located in the plasma membrane unless otherwise indicated. NA, nicotianamine.

Morales de los Ríos *et al.*, *New Phytologist*, CC BY 4.0.

DOI:10.1111/nph.70730

NPF family: clade partitioning

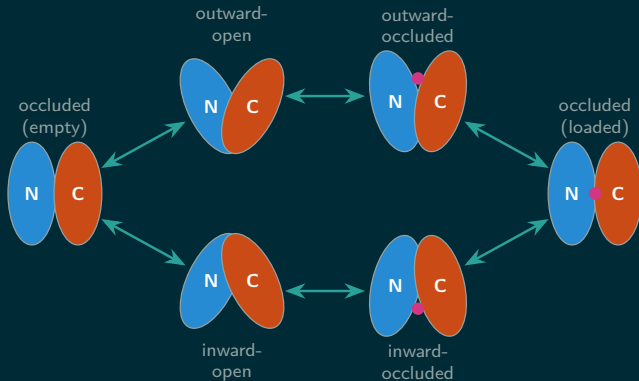
- 53 NPF sequences in *Arabidopsis thaliana*, aligned with MUSCLE
 - Tree splits the family into 8 clades (NPF1–NPF8 subfamily numbering)
 - Highlighted in red: the best-characterized importers – NPF6.3/NRT1.1, NPF6.2/NRT1.4, NPF7.3/NRT1.5
- ⇒ Clade position gives a natural axis for comparing predicted conformations



Phylogenetic tree of the NPF family (*A. thaliana*), 53 sequences aligned with MUSCLE; NPF6.3/NRT1.1, NPF6.2/NRT1.4 and NPF7.3/NRT1.5 marked in red. Morales de los Ríos *et al.*, *Plant Physiology and Biochemistry* 168 (2021),

MFS transport cycle: six conformations

Alternating-access mechanism: transport requires passing through a fully **occluded** intermediate in either direction – the **biological reason** the pipeline targets six conformations per protein, not six independent opinions on one.



What's experimentally known

- Only **2 NPFs** have experimentally resolved structures:
 - NPF6.3 (4OH3, 5A2N, 5A2O)
 - NPF2.10 (9UI1, 9UI6, 9UIF, 9UIT)
- Only **importers** are well characterized in the literature (technical constraints)
- **Exporters** remain unidentified



NPF6.3 – PDB 4OH3, X-ray, 3.25 Å

Sun *et al.*, *Nature* 2014

Research project goals

1. Ab initio modelling of 3D structures

1.1 Pipeline to generate the six conformations

(this project)

1.2 Network interaction analysis (ligand and intra-chain)

2. Functional annotation of NPFs

2.1 Identify key features of importers vs. non-importers

2.2 Eventually, find exporters

The question

- Arabidopsis **NPF** (Nitrate/Peptide Transporter) family

Which conformations can each NPF transporter adopt, apo and ligand-bound?

- Structure prediction as a proxy for conformational sampling

Why not just one model?

- Sibling repo AF3_NPF_pipeline already sampled **30 AF3 seeds/protein**
- Overlaying those ensembles with Boltz-2 (Gibberellin importers, 8 proteins) showed: AF3 alone **does not** recover every conformation a second, architecturally-different model finds

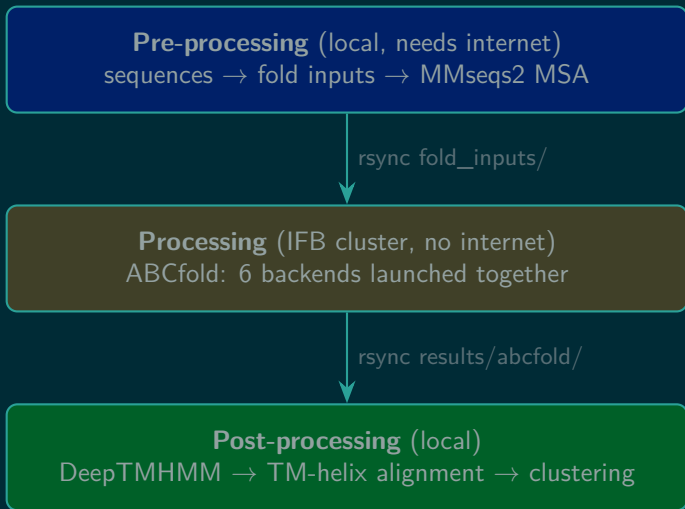
⇒ One model isn't enough – run **six together** instead of resampling one

The six backends

- AlphaFold3
- Boltz-2
- Chai-1
- OpenFold3
- Protenix
- RosettaFold3

All launched **together** per protein via ABCfold's `abcfold -abcopr` against the *same* resolved fold input.

Three stages



Stage 1–3: from sequence to resolved fold input

1. **Sequences** – UniProt → per-protein FASTA
2. **Fold input** – `fold_input.json` per protein × apo/holo (AF3-dialect JSON, ABCfold's native input: sequence + every replica seed)
3. **MMseqs2 default run** – MSA + top-hit templates from the ColabFold webserver, no manual curation, no pocket restraint; embedded into `fold_input.resolved.json`

Runs locally, driven by Snakemake (`workflows/preprocessing/Snakefile`); resumable – existing resolved inputs are never re-fetched.

Stage 4: ABCfold on the IFB cluster

- One `abcfold -abcopr...` call per protein $\times$ form
- Launches all 6 backends against the same `fold_input.resolved.json`
- AlphaFold3 backend runs via IFB's `alphafold` module (Singularity wrapper, auto-discovered) – no separate image needed
- Boltz/Chai-1/OpenFold3/Protenix/RosettaFold3 built once via `--prime` on a login node (needs internet)

Gibberellin importers run first

- 8 GA1-ligand proteins (NPF2.1, NPF2.5, NPF2.10, NPF2.12, NPF2.13, NPF3.1, NPF4.1, NPF5.6) – the group already studied with AF3 + Boltz-2
- Listed in `priority_gibberellin.txt`, put first in the SLURM array manifest
- Fastest path to validating the six-backend approach against a known result

Stage 5–6: topology and alignment

1. **DeepTMHMM** – TM-helix topology (BioLib), once
2. **TM-helix Kabsch alignment** – per protein $\times$ form, pooled across every backend and seed

Output: `results/tm_alignment/<protein>/{aligned_ca_tm.npy, meta.csv}` – `meta.csv` now carries a `model` column (`alphafold3/boltz/chai1/openfold3/protenix/rosettafold3`).

Stage 7: clustering & visualisation

- Notebook-based (JupyterLab, envs/notebook.yaml)
- Same schema as AF3_NPF_pipeline's clustering notebooks + one new axis: **color/facet by backend**
- *That's the whole point of this pipeline:* which backend(s) actually reach which conformational cluster?
- Status:

Where we are



Next steps

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Questions?